Finite-dimensional subcase of the positive commutator problem

Literature-implied status note

2026-06-18

Status. `literature_implied_answer (partial subcase)`.

Source problem. Madou, Remizov, and Vafadar, *Open problems in one-parameter operator semigroups theory*, arXiv:2410.00416, Problem "Positive commutator problem" asks: if E is a Banach lattice and $C : E \to E$ is a positive quasinilpotent compact operator, do there exist positive operators $A, B : E \to E$ such that

$$C = AB - BA$$

with one of A and B compact?

Supporting theorem. Drnovsek and Kandic, *Positive operators as commutators of positive operators*, arXiv:1707.00882, Corollary 3.2, prove that a positive matrix C can be written as a commutator of positive matrices A and B if and only if C is nilpotent. Moreover, their construction can choose A diagonal and B permutation-similar to a strictly upper triangular matrix.

Identification. Let E be a finite-dimensional Banach lattice. Then E is lattice isomorphic to $\mathbb{R}^n$ with its coordinate order, so positive operators on E correspond to positive matrices. Every operator on E is compact. Finally, in finite dimension quasinilpotence is equivalent to nilpotence. Therefore, if $C : E \to E$ is positive, quasinilpotent, and compact, then C is a positive nilpotent matrix in lattice coordinates. By Drnovsek–Kandic Corollary 3.2, $C = AB - BA$ for positive matrices A, B, and since the space is finite-dimensional one of them is automatically compact.

Scope and provenance. This answers only the finite-dimensional Banach-lattice subcase of the source problem. The supporting paper predates arXiv:2410.00416 and does not present itself as answering that later problem; the relation is an agent-identified specialization. The general infinite-dimensional compact/quasinilpotent Banach-lattice problem is not settled by this packet.

References.

- K. R. Madou, I. Remizov, and R. Vafadar, *Open problems in one-parameter operator semigroups theory*, arXiv:2410.00416.
- R. Drnovsek and M. Kandic, *Positive operators as commutators of positive operators*, arXiv:1707.00882; Studia Math. 245 (2019), 185–200.